

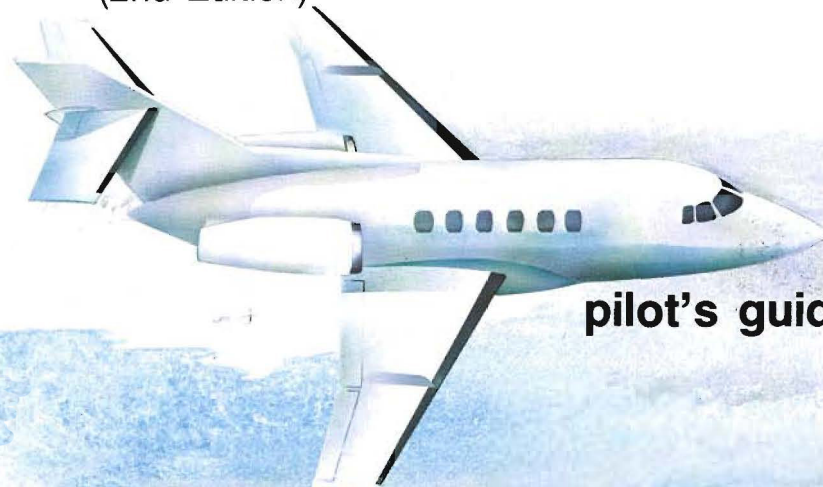


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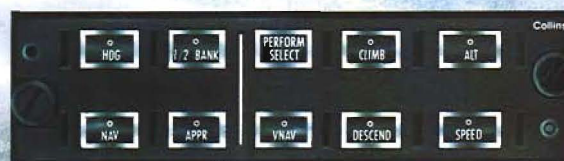
Collins General Aviation Division

Collins APS-85 Autopilot System (For the BAe 125-800)

(2nd Edition)



pilot's guide



Avionics Group/Rockwell International
Cedar Rapids, Iowa 52498

MODE	PARAMETER	VALUE
HDG	ROLL LIMIT ROLL RATE CMD FILTER ACCURACY	27° 5°/SEC + 1 SEC LO PASS 0.5° RELATIVE TO HEADING REFERENCE
NAV CAP	INTERCEPT LIMIT ROLL LIMIT ROLL RATE CMD FILTER	±90° 27° 5°/SEC + 1 SEC LO PASS
ON COURSE	CROSSWIND CORRECTION ACCURACY	±30° 15 μA MAX WITH NO STEADY STATE OSCILLATIONS
APPR LOC CAP	INTERCEPT ANGLE OVERSHOOT	±90° 1 DOT MAX, LESS THAN 25 μA WHEN NOT GEOMETRY LIMITED
LOC ON COURSE	ROLL LIMIT CROSSWIND CORRECTION ACCURACY	±15° ±30° CAT II LIMITS (±20 μA)
GS	OVERSHOOT ACCURACY	1 DOT MAX, 75 μA MAX CAT II LIMITS (±65 μA)
VS	PITCH CMD LIMIT MIN CMD INCREMENT ENGAGE RANGE MAX DEVIATION	0.15 g MAX OR PITCH RATE PROGRAMMED WITH TAS 50 FT/MIN UP TO ±6000 FT/MIN ±600 FPM WITH CONFIG CHANGE
IAS	PITCH CMD LIMIT RANGE MAX DEVIATION MIN MCD INCREMENT	0.15 g 60 TO 500 KT ±5 KT, NONMANEUVER 2 KT
ALT	ALT ENGAGE RANGE OVERSHOOT MAX DEVIATION CONFIGURATION CHANGE	-1000 TO +55 000 FT APPROX 10% OF VS UP TO 2000 FT/MIN ±50 FT OR ±0.12% OF ALT, WHICHEVER IS GREATER ±100 FT OR ±0.4% OF ALT
ALT SELECT	CAPTURE RANGE VS RANGE CAPTURE G OVERSHOOT	0 TO 50 000 FT 6 000 FT/MIN 0.2 MAX 100 FT MAX

*The asterisk indicates pages changed, added, or deleted by the current change.

Title.....1 Sep 88
List of Effective Pages.....1 Sep 88
i thru ii1 Sep 88
1 thru 241 Sep 88

RETAIN THIS RECORD IN THE FRONT OF MANUAL.
ON RECEIPT OF REVISIONS, INSERT REVISED PAGES IN THE
MANUAL, AND ENTER DATE INSERTED AND INITIALS.

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ACTION	RESPONSE
Release EFIS GROUND TEST button.	When the self-test switch is released, the displays will return to normal operation. An external installer-supplied comparator reset button or switch must be actuated to reset the PIT, ROL, and HDG comparator warning messages. Actuating master reset once stops the PIT, ROL or HDG indications from flashing. Actuating comparator reset again will eliminate the comparator messages if the condition causing the comparator warning has been eliminated.
NOTE The position of the elevator trim tab should be checked after the autopilot is disengaged to ensure that the tab is at the desired position.	

ACTION	RESPONSE
Push the AP DISC button	AP annunciation on PFD will become red and may flash until pilot acknowledgement. Refer to the Flight Manual Supplement for the correct operation.
Apply force to the primary controls.	Controls do not resist movement.
Select HDG mode on the mode select panel.	PFD command bars come into view.
Press and hold EFIS GROUND TEST button.	When self-test is first initiated, an increment of 10 degrees is added to the current values of pitch and roll, and 20 degrees is added to heading. A positive increment is added to the pilot's side and a negative increment is added to the copilot's side. The word TEST, in red, will appear on the PFD and the MFD while the test is in process. The addition of the differential 10- and 20-degree increments causes comparator logic in the EFIS system to give flashing PIT (pitch), ROL (roll), and HDG (heading) comparator warning messages. A copyright message also appears during test. The Category II comparator and excessive deviation warnings may also be tested. If the Category II select switch is actuated and the on-side localizer is tuned, the Category II warning annunciators are displayed. These include GS compare, LOC compare, RA compare glideslope excessive deviation, and localizer excessive deviation. If the self-test switch is held for longer than 4 seconds, the pitch, roll, and heading increments are removed from view and all flags associated with the PFD and ND displays will be brought into view.

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Preflight Checklist

The following items provide a limited check on availability of the autopilot and the EFIS. It is recommended that the items be checked before every takeoff, especially before the first flight of each day and after any maintenance activity. (Any maintenance activity directly involving the flight control system should be followed by a comprehensive ground check of the system.)

ACTION	RESPONSE
Apply airplane power to all units in the autopilot system, EFIS, air data system, and attitude/heading system. Delay at least 70 seconds to allow attitude/heading system to initialize. Do not move the airplane during this initialization period.	ATT and HDG flags are out of view and EFIS indicates correct airplane magnetic heading and pitch/roll attitude.
Press FLT ANNUN TEST button	All mode select panel and autopilot panel button annunciators illuminate, and all autopilot associated external annunciators illuminate.
Engage the autopilot and yaw damper and, in turn, apply force to the control yoke, control column, and rudder pedals.	Primary controls resist movement and elevator trim operates in the appropriate direction.
NOTE	
The diagnostics mode should be entered if the system will not engage during the preflight checks. Refer to the DIAGNOSTICS section of this pilot's guide, page 19, for detailed instructions.	
Move autopilot panel pitch wheel and turn knob.	Control column and yoke follow inputs.
Using EFIS system HDG knob, set ND heading cursor under lubber line, select HDG mode on the mode select panel, and move the heading cursor.	PFD command bars come into view and the control yoke follows the heading cursor inputs.

Procedure for Recording the Diagnostic Codes

- a. Enter APS-85 diagnostics by first pressing and holding the installer supplied EFIS GROUND TEST button and then pressing and releasing the radio altimeter test (TST) button on the DSP (display select panel for EFIS system). Release the EFIS GROUND TEST button.
- b. Press and release the radio altimeter test (TST) button three (3) more times (FCS DIAGNOSTICS should be displayed at the top of the PFD).
- c. Access APS-85 diagnostics by simultaneously pressing any three MSP mode select buttons and holding them in for approximately three seconds. Release the buttons.
- d. Record any REPAIR, AP DIS, AP ENG, YD DIS, and/or YD ENG CODE displayed on the PFD that contains digits other than zero. (If all zeros, no errors have been detected.) Indicate whether the CODE is from the LEFT (pilot's) side or RIGHT (copilot's) side. Give the recorded CODE(s) to maintenance personnel for interpretation and corrective action.
- e. Exit diagnostics by first pressing and releasing the radio altimeter test (TST) button on the DSP, then press and release the installer supplied EFIS comparator reset button to clear the flashing comparator warnings displayed on the PFD.
- f. The word TEST, in yellow letters, should have appeared in the upper right corner of the PFD after the radio altimeter test (TST) button was pressed. This is to inform the pilot that the APS-85 is still in the diagnostic mode even though the EFIS displays have returned to normal.
- g. Press and release the AP DISC (autopilot disconnect) button. Verify that the word TEST disappears from the upper right corner of the PFD. The APS-85 is now out of the diagnostics mode and normal operation is restored.
- h. Resume operation or remove power from the system if ready to shut down at the end of the flight.

Introduction

This pilot's guide presents operating information about the Collins APS-85 Autopilot/Flight Guidance System. The general operational information contained in this pilot's guide must be supplemented with detailed information contained in the BAe-800 Airplane Flight Manual which describes operation of the system as installed in a specific airplane.

The APS-85 is integrated with a digital EFIS-85 Electronic Flight Instrument System, an ADS-82 Air Data System, and an AHS-85 Attitude Heading System to become an FCS-85 Flight Control System. Additional data is required from various other airplane systems such as NAV receivers, DME, RNAV, INS, VLF, LNV, etc.

The APS-85 is a digital autopilot system designed for automatic control of high-performance airplanes. When it is engaged, the autopilot maneuvers the airplane and the pilot monitors the flight path by observing information displayed by the flight instrument system. When the autopilot is disengaged, the pilot manually flies the airplane in response to the PFD command bars.

Whether the autopilot is engaged or disengaged, the flight control system provides commands to:

- Hold a desired attitude
- Maintain a barometric altitude
- Hold a vertical speed
- Hold a Mach number
- Hold an indicated airspeed
- Capture and maintain a preselected altitude
- Capture and maintain a preselected VNAV path
- Capture and track a preselected heading
- Capture and track a preselected radio course (VOR, LOC, GS)
- Capture and track a lateral navigation course (INS, VLF, LNV, RNV, etc)
- Capture and track a localizer and glideslope to established Category II minimums
- Maintain a wings-level, fixed pitch-up attitude for go-around

Before Using The System

Read Thoroughly

The APS-85 Autopilot System is designed to provide automatic flight control of high-performance airplanes. Some components of the system require adjustment to match specific airplane characteristics or to meet specific operational requirements. The following statements should be read carefully and understood by the pilot before using the system.

- a. Installation of the correct equipment is required to ensure correct operation. Interlocks are provided for critical functions in the system. However, the user must ensure that equipment installed is the correct type and has been correctly adjusted and tested. Refer to the Airplane Flight Manual and other information provided with the airplane to determine the exact equipment complement required.
- b. The APS-85 is intended for use as a flight control system ONLY as certified on a specific airplane. Modifications can cause incorrect operation and may invalidate the certification of the system. All modifications should be made by a certified modification center that has authority to change the airplane certification.
- c. Each FAA approved Airplane Flight Manual contains detailed data on the operation of the APS-85 in that specific airplane. The Airplane Flight Manual should be used to determine any variations in operation of the system from the general operation described in this pilot's guide.
- d. As a result of the sophistication of the airplane, the flight control system, and the peripheral avionics equipment, an almost unlimited number of combinations of conditions could occur which could result in the APS-85 not operating precisely as described. Examples of such conditions include, but are NOT limited to, the following:

Diagnostics

This section provides the procedure for accessing the diagnostic mode of the APS-85 Autopilot System. As an aid to maintenance personnel, the pilot should access the APS-85 diagnostics and record any REPAIR, AP ENG, and/or YD ENG CODE displayed on the PFD in the EFIS system if the system will not engage during the preflight check. The diagnostics should also be entered at the end of each flight if a failure occurs and any REPAIR, AP DIS, and/or YD DIS CODE should be recorded. This must be done **BEFORE** power is removed from the system since flight data is not retained after power is removed from the flight control computer. Any diagnostic codes that are recorded should be given to maintenance personnel for further troubleshooting action.

Conditions For Entering Diagnostics

The diagnostics mode can be entered whenever power is applied to the system, the autopilot and yaw damper are disengaged, the airplane is on the ground, and the EFIS system is in that portion of its test routine that allows diagnostics from the APS-85 to be displayed on the PFD.

When the APS-85 diagnostics is first entered, the REPORT MODE is automatically selected.

Report Mode

The REPORT MODE displays the status of selected parameters internal to the flight control computer's software. The following is an example of the initial display on the PFD when REPORT MODE is selected. Note that in the example a REPAIR CODE of 0008C0 is present in the LEFT (pilot's side) column.

REPORT MODE

	LEFT	RIGHT
REPAIR CODE	= 0008C0	000000
AP DIS CODE	= 000000	000000
AP ENG CODE	= 000000	000000
YD DIS CODE	= 000000	000000
YD ENG CODE	= 000000	000000

GO-AROUND

- Controlled by GA buttons on control yokes.
- Selecting GA causes the AP to disengage, cancels selected lateral and vertical modes, and provides wings-level, fixed pitch-up commands.
- Push the GA button a second time to cancel the red flashing AP message on the PFD.
- GA mode can be selected at any time and should be selected at the decision to discontinue the approach.
- AP can be reengaged after becoming established in GA attitude (basic mode will occur and the PFD command bars will remain in view, but the ND heading cursor will not synchronize).
- Airplane heading at moment of GA selection is held throughout GA (not the heading shown by the heading cursor on the ND).
- Select any new vertical mode to clear vertical portion of GA mode.
- Select any new lateral mode to clear lateral portion of GA mode.

Trim System

The trim system is an electrically driven servo that off-loads forces on the elevator. The trim system is automatically controlled by the flight control computer when the autopilot is engaged. When the autopilot is disengaged, the pilot operates the system manually with the elevator trim switches.

A preengage test, performed by the autopilot, includes a check of the trim system. If a failure is detected in the trim system, the autopilot will not engage. If a trim failure occurs after the autopilot is engaged, the trim fail annunciator turns on and only the trim servo is disengaged.

The flight control computer continuously checks for mistrim conditions when the autopilot is engaged. The computer turns on the appropriate mistrim annunciator when forces on the flight control surfaces exceed normal levels. Commands are automatically sent to the trim system to correct elevator mistrim. Aileron mistrim must be corrected manually with the aileron trim wheel. Mistrim annunciators turn off when the mistrim conditions are removed.

1. Power failure or conditions where the power does not meet the applicable specifications. This includes loss of power, an open fuse or circuit breaker, a power switch off, a fault in the system, or power voltage or frequency out of specification limits.
 2. Use of the system when warning flags or annunciations indicate that a problem is present.
 3. Operation of the system controls incorrectly or in the improper sequence.
 4. Installation of equipment not certified for use on a specific airplane.
- e. The APS-85 Autopilot System has been designed to exhibit a very high degree of functional integrity. The monitoring, diagnostics, and self-test capabilities of the system detect more system failures than previous systems. However, the pilot is reminded that even though an undetected failure condition will be very unusual, failure detection is not 100 percent and the pilot should continue using standard instrument cross-check procedures in the unlikely event that an undetected failure does occur.

If any of the above or similar conditions exist, adjust operation to correct the problem or use emergency procedures as required.

NOTE

As an aid to maintenance personnel, the pilot should take advantage of the diagnostic capabilities designed into the APS-85 Autopilot System. The diagnostics mode should be entered if the system will not engage during the preflight check, and at the end of each flight BEFORE power is removed from the system. Refer to the DIAGNOSTICS section of this pilot's guide, page 19, for detailed instructions.

The FCS-85 Flight Control System

The FCS-85 Flight Control System is a combination of autopilot, guidance, displays, and sensors. The following paragraphs provide brief descriptions of the units that are used in the system.

Flight Control Computer (FCC)

The flight control computer uses magnetic heading, radio data, acceleration, attitude, mode select panel, autopilot panel, autopilot servos, and air data inputs to develop lateral and vertical commands for the autopilot and for the flight instrument system command bars or crosspointers. This single computer concept allows all of the mode operations in the flight director to be flown automatically by the autopilot. Completely dual digital computation channels and dual servo amplifiers provide 3-axis fail-passive attitude control. The dual computation channels ensure that the copilot's flight guidance functions are isolated from the pilot's flight guidance.

Autopilot Panel (APP)

The autopilot panel is the cockpit control center for the autopilot function of the system. The autopilot panel contains both autopilot and yaw damper engage/disengage levers, optional autopilot transfer and turbulence mode select buttons, and the pitch wheel/turn knob assembly.



- Selects speed mode.
- Can be used with any lateral mode.
- If selected below airplane's MACH transition altitude, then IAS mode occurs.
- If selected above airplane's MACH transition altitude, then MACH mode occurs.
- Once SPEED mode is selected, press PERFORM SELECT button to alternate between IAS and MACH mode.
- Annunciated in green on PFD with numbers indicating reference IAS or MACH.
- Reference IAS or MACH can be modified by rotating the pitch wheel or for IAS only, by the airspeed bug on the airspeed indicator.



- Multifunction button provides the following functions:
 - In CLIMB mode — Selects L (low), M (mid), or H (high) speed climb profile.
 - In ALT or ALTS mode — Displays or removes cruise advisory airspeeds on the PFD.
 - In SPEED mode — Alternates between IAS and MACH mode.

Special modes

VERTICAL SYNC (flight director only)

- Controlled by SYNC button on control yoke.
- Can be used with any vertical mode except during GS operation.
- Allows changing reference value of a vertical mode without reselecting the mode.
- Press the SYNC button, fly the airplane to the new reference value, and then release the SYNC button.
- Press the SYNC button to cancel the vertical portion of GA mode and sync the system to the existing pitch attitude.

ALTS

- Altitude preselect mode (does not have a mode select button).
- Mode is automatically armed whenever preselector/alerter setting differs from indicated altitude by ± 250 ft (except after GS capture).
- Can be used with any lateral mode.
- 0 to 50 000 ft capture range.
- $\pm 6\,000$ ft/min VS range.
- Capture g's 0.2 max.
- 100 ft max overshoot.
- SSEC and BARO settings are included in altitude preselect computations (pilot must adjust power levers to obtain desired airplane performance during the capture maneuver).
- Deviation from preselected altitude displayed on right side of PFD.
- System goes to VS and rearms for ALTS if a new altitude is dialed in on the preselector/alerter during altitude preselect capture (prior to track).
- System goes to ALT hold and rearms for ALTS if a new altitude is dialed in on the preselector/alerter after altitude preselect track.
- Annunciated in green on PFD with cruise advisory airspeeds as described for ALT mode, if selected.
- Rotate pitch wheel to 'take command' and go to VS mode.



(optional)

- Selects vertical navigation mode.
- Can be used with any lateral mode.
- Define VNAV path on VNI-80 — after capture, system tracks that path.
- Deviation from VNAV path displayed on right side of PFD.
- Annunciated in green on PFD.
- Rotate pitch wheel to 'take command' and go to VS mode.



- Selects descend mode.
- Can be used with any lateral mode.
- Gives initial 1 000 ft/min rate of descent.
- Annunciated in green on PFD with numbers showing ft/min down.
- Rotate pitch wheel or VS bug knob to change vertical speed reference (only negative values are valid).

Mode Select Panel (MSP)

The mode select panel is used to select the lateral and vertical modes of operation for the flight control system.

Primary Servos and Servo Mounts

The primary servos position the airplane control surfaces in response to commands from the flight control computer.

Trim System

The trim system is an electrically driven servo that off-loads forces on the elevator.

Other System Components

An EFIS system is considered part of the FCS-85. EFIS components installed in the cockpit and used by the pilot include the PFD (primary flight display), the ND (navigation display), the MFD (multifunction display), and associated system controls. Refer to the applicable EFIS system pilot's guide for descriptions, display formats, operating information, etc. Also part of the FCS-85 is the Collins digital ADS-82 Air Data System and the Collins AHS-85 Attitude Heading System. Inputs to the system from various sensors such as navigation receivers, DME, LNV, RNAV, FMS, etc are also required. In addition, various installer supplied items such as remote annunciator panels, reversionary switching provisions, etc are provided. Refer to the Airplane Flight Manual Supplement for operational details.

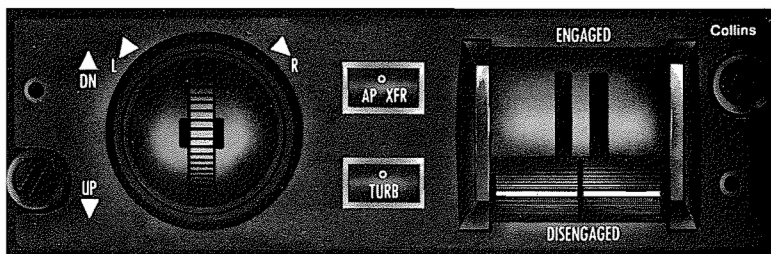
The APS-85 Autopilot System

CAUTION

During autopilot and flight director operations, a pilot must be seated at the controls with seat belt fastened.

The autopilot may be engaged anywhere in the flight envelope during normal airplane attitudes. In the BAe-800 installation, yaw damper engagement is independent of autopilot engagement, but engaging the autopilot automatically engages the yaw damper.

APP-85() Autopilot Panel



The APP-85() Autopilot Panel contains the turn knob and pitch wheel for manually controlling the system. The AP XFR and TURB buttons are push-on/push-off (alternate action) momentary buttons used to select autopilot transfer and turbulence modes. Indicators in the buttons annunciate the mode condition of the system. (In the following descriptions, the buttons on the APP are shown boxed.) The autopilot and yaw damper are engaged by moving the guarded engage levers to the up position. These levers are mechanically connected to the engage switches and provide positive engage and disengagement of the system.

- In VS mode, the VS reference is synchronized to the airplane's vertical rate.
- VS may be modified by either the pitch wheel or by the bug on the VSI/VNI.
- Annunciated in green on PFD with numbers showing ft/min and an upward or downward pointing arrow showing VS direction.
- Rotate pitch wheel to 'take command' and go to VS mode with VS reference synchronized.



- Selects climb mode.
- Can be used with any lateral mode.
- Flies a preprogrammed IAS/MACH profile that is matched to the performance characteristics for this airplane.
- Press the PERFORM SELECT button to select one of three speed profiles:
 - L for low speed, 230 kt/.63 MACH
 - M for mid speed, 250 kt/.63 MACH
 - H for high speed, 250 - 260 kt/.65 MACH
- Speed profiles may be optimized for minimum fuel climb, best angle climb, or maximum rate of climb.
- Annunciated in green on PFD with numbers followed by L, M, or H indicating preprogrammed IAS/MACH profile.
- Rotate pitch wheel to 'take command' and go to VS mode.



- Selects altitude hold mode.
- Can be used with any lateral mode.
- -1 000 to +55 000 ft engage range.
- ±1 000 ft/min VS range.
- SSEC is included in computations but BARO settings are not.
- Press the PERFORM SELECT button to display cruise advisory airspeeds displayed on the PFD.
- Annunciated in green on PFD.
- Rotate pitch wheel to 'take command' and go to VS mode.

Vertical modes

PITCH (hold)

- Flight director only mode (does not have a mode select button).
- Can be used with any lateral mode.
- Mode occurs when AP is disengaged and no other vertical mode is selected and a lateral mode is selected (PFD command bars are removed unless a lateral mode is selected).
- Commands are to maintain the pitch attitude existing when pitch hold mode occurs.
- Pitch attitude is changed by manually flying the airplane to the desired pitch attitude, then press and release the SYNC button.
- Annunciated in green on PFD.

PITCH WHEEL

- APP pitch wheel functions as vertical reference command knob so that IAS, VS, and MACH command capability is standard.
- Can be used with any lateral mode.
- 0.15 g max pitch command limit.
- Rotate the pitch wheel and observe the PFD command bars to change pitch attitude.
- Reference bugs of on-side air data instruments will track the pitch wheel slew command.
- Reference bugs of on-side air data instruments can also perform the command function.
- Rotate the pitch wheel to 'take command' and go to VS mode when system is in CLIMB, DESCEND, ALT hold, altitude track, or VNAV mode.
- Pitch wheel is inoperative after GS capture, during altitude capture, or when AP is disengaged.

VS

- Basic vertical mode of the system (does not have a mode select button).
- Mode occurs when AP is engaged and no other vertical mode is selected.
- Can be used with any lateral mode.
- 0.15 g max pitch command limit.
- 50 ft/min minimum command increment.
- Up to $\pm 6\,000$ ft/min engage range.

TURN Knob

- With AP engaged, rotate turn knob to 'take command' of aileron channel and go to basic mode (roll hold).
- Can be used with any lateral mode except NAV or APPR after localizer capture has occurred.
- Roll rate is proportional to knob displacement.
- 5°/sec roll rate limit.
- 32° roll limit.
- With bank angle more than 5° when knob is returned to detent, system holds bank angle. With bank angle less than 5°, system goes to heading hold.
- Turn knob is inoperative when localizer is captured and also when AP is disengaged.

PITCH Wheel

- With AP engaged, rotate pitch wheel to 'take command' of elevator channel, go to basic mode (VS) if system is in CLIMB, ALT hold, altitude track, or VNAV mode, and slew the vertical speed reference indicated by the reference bugs of the on-side air data instruments.
- Reference bugs of on-side air data instruments can also perform the command function.
- Pitch wheel functions as vertical reference command knob so that DESCEND, IAS, VS, and MACH command capability is standard.
- Pitch rate is proportional to pitch wheel displacement.
- Pitch rate command is programmed with TAS resulting in equal pitch wheel displacement giving same g loads (0.15 g max pitch command limit) throughout entire flight regime.
- Pitch wheel is inoperative after vertical capture in APPR mode, during altitude capture, or when AP is disengaged.



- Selects turbulence mode.
- Button is push-on/push-off and when selected, reduces AP gains to provide proper lateral and vertical control in turbulent conditions.
- An indicator in the TURB button illuminates when mode is selected.
- Mode is automatically cleared and inhibited by localizer capture.



- Selects autopilot transfer.
- Button is push-on/push-off and when selected, transfers AP signals and monitor inputs to copilot's system.
- Vertical commands, other than VNAV or vertical capture in APPR, synchronize with AP XFR.
- A second push of the button transfers AP command inputs back to pilot's system.
- An indicator in the AP XFR button illuminates when transfer has occurred.

AP and YD engage levers

- Dual section levers with two positions, ENGAGED and DISENGAGED.
- Left-hand lever is for YD (yaw damper), right-hand lever is for AP (autopilot).
- Levers are mechanically and electrically interlocked allowing the YD to be engaged independently from the AP but ensuring that if the AP is engaged the YD will also be engaged.

Autopilot self-test

- A self-test sequence is initiated each time the AP is powered up (test takes approximately 60 seconds to complete).
- The system will engage only if the power-up test sequence is completed successfully and all interlocks are correct.
- If a monitored fault exists, the AP will not engage and an 'FD' flag will appear in the lower left center of the PFD.
- The AP will not engage if the equipment or the airplane is improperly configured.

NOTE

The diagnostics mode should be entered if the system will not engage. Refer to the DIAGNOSTICS section of this pilot's guide, page 19, for detailed instructions.



- Selects navigation mode.
- Can be used with any vertical mode.
- 5°/sec roll rate limit.
- 15° roll limit for LOC on course, 27° roll limit for VOR, TCN, LNV, RNV, etc.
- Provides automatic crosswind correction with crab angles up to 30°.
- Provides NAV capture angles up to 90°.
- Adaptive captures include TAS compensation of the capture points.
- NAV system to be flown is the one selected for display by the EFIS.
- AP always flies the primary course being displayed on the ND
- System operates in HDG mode prior to NAV capture.
- Prior to capture, NAV source is annunciated on the PFD in white; after capture, annunciation is green.
- When in VOR capture, the system automatically reverts to HDG mode and LOC arm by tuning the active NAV receiver to a LOC frequency.
- Rotate APP turn knob to 'take command' and go to roll hold mode.



- Selects approach mode.
- Can be used with any vertical mode prior to GS capture.
- 5°/sec roll rate limit.
- 15° roll limit for LOC on course.
- Provides automatic crosswind correction with crab angles up to 30°.
- Provides NAV capture angles up to 90°.
- Approach type to be flown (ILS, B/L, VOR, RNAV, etc) will always correspond with the data selected for the primary course being displayed on the ND.
- Lateral capture and annunciation are the same as for NAV mode.
- For an ILS approach, GS is armed after LOC capture.
- At GS capture, the vertical mode in use automatically clears.
- System provides superior path tracking and damping for the LOC down to CAT II minimums.
- Automatically senses back-course conditions and provides proper signal reversing.
- APP turn knob is inoperative after lateral capture.

Lateral modes

ROLL (hold)

- Basic lateral mode of the system (does not have a mode select button).
- Mode occurs automatically when:
 - System is initially powered up.
 - No other lateral mode is selected.
 - AP is reengaged following selection of GA mode.
 - AP is engaged and APP turn knob is rotated (except in APPR mode after lateral capture).
- 5°/sec roll rate limit.
- 32° roll limit.
- Annunciated in green on PFD.
- PFD command bars are removed from view in roll hold mode.
- With AP engaged, system holds bank angle if angle is more than 5°. With bank angle less than 5°, system goes to heading hold.



- Selects heading mode.
- Can be used with any vertical mode.
- 5°/sec roll rate limit.
- 27° roll limit.
- Annunciated in green on PFD.
- Rotate APP turn knob to ‘take command’ and go to roll hold mode.



- Used with HDG and NAV modes.
- 12.5° roll limit.
- Automatically clears by selecting APPR mode or after LOC capture.

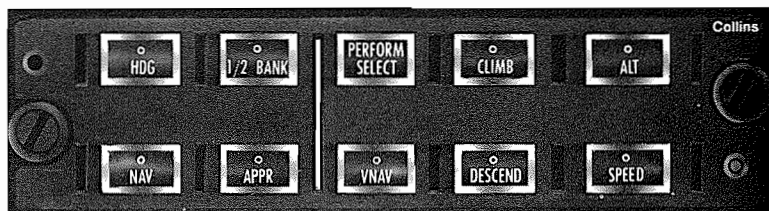
Nonengage and disengage conditions of the APS-85 Autopilot System are summarized in the following table.

MANUAL DISENGAGE	AUTOMATIC DISENGAGE	FAILURE TO ENGAGE
Press AP disengage button on either control yoke	Select GA mode	Open disengage button
	Fault detected in servo-loop circuits	Any of the automatic disengage criteria
	Loss of attitude system valid	
Move AP and YD levers to DISENGAGED position	Loss of system power	Failure to pass power-up test
	Fault in engage circuits	Airplane incorrectly configured
	Loss of yaw damper valids	Flight control computer incorrectly configured
NOTE		
There may be additional interlocks unique to a specific airplane that inhibit engagement or cause automatic disengagement of the autopilot. Refer to the BAe-800 Airplane Manual for detailed information concerning autopilot operation specific to the airplane.		

Autopilot and yaw damper annunciation

The autopilot/yaw damper annunciations are located on the third and fourth lines to the left of center in the upper part of the PFD. The autopilot and yaw damper engaged annunciation is AP and YD. If a disengagement occurs (for either the autopilot or the yaw damper), the engage messages turn red and flash until acknowledged by the flight crew (usually a second push of the AP disconnect button). When the acknowledgement occurs, the message disappears.

MSP-85() Mode Select Panel



All computed modes for the APS-85 system are selected on the MSP-85() Mode Select Panel. A solid vertical divider strip separates the four lateral modes on the left side of the MSP (two upper and two lower buttons) from the six vertical mode buttons on the right side of the MSP (three upper and three lower buttons). Push-on/push-off momentary pushbuttons are provided for the modes. The PERFORM SELECT button is not considered a mode button; this function is used in conjunction with the CLIMB, SPEED, ALT hold, and altitude preselect modes. Refer to the Airplane Flight Manual Supplement for further information.

Pressing the pushbutton for a mode that is not selected clears all noncompatible modes and causes the new mode to latch if all conditions for the mode are satisfied. An annunciator light in the mode button illuminates if all conditions for the mode are satisfied. Flight control mode annunciators are located on the top left and right sides of the EFIS PFD. The vertical mode annunciations are located on the first two lines to the left of center and the lateral modes are located on the top line to the right of center. The active vertical and lateral modes are shown in green and are closest to the center of the PFD with the armed modes and submodes in white outboard of the active modes. When a new active mode is displayed, either by being selected or automatically capturing, it will flash for 5 seconds before becoming steady. When no lateral mode is selected, roll hold (basic lateral mode) results and the PFD command bars are removed from view. When no vertical mode is selected and the autopilot is not engaged, pitch hold mode is automatically selected. When no vertical mode is selected and the autopilot is engaged, VS (basic vertical mode) results. Selection of a lateral mode will bring the PFD command bars into view. With the autopilot engaged and no mode selected on the MSP, the autopilot accepts pitch and turn commands from the pitch wheel and turn knob on the APP.

In addition to the mode select buttons on the MSP, switches on the control yoke provide selection of vertical sync and go-around mode.

The system initially powers up with roll hold (basic lateral mode) and pitch hold modes automatically selected (the autopilot will be disengaged at initial power up). With the autopilot engaged, the system will revert to the roll hold mode when the turn knob is rotated (except in APPR mode after lateral capture), and to VS hold mode (basic vertical mode) when the pitch wheel is rotated (except in APPR mode after vertical capture, or during altitude capture). The system also reverts to the basic modes whenever other modes release, after power is momentarily interrupted, or after the autopilot is reengaged following selection of go-around mode.

In the following descriptions, mode buttons on the MSP are shown boxed.